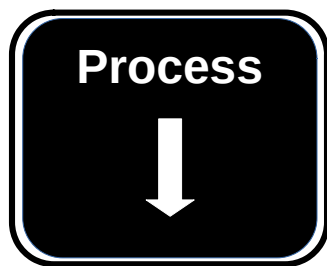


Fix: externalizing network I/O in serverless computing

Yuhan Deng, Akshay Srivatsan, Sebastian Ingino, Francis Chua, Yasmine Mitchell, Matthew Vilaysack,
Keith Winstein
Stanford University

“It has been said that the principal function of an operating system is to define a number of different names for the same object, so that it can busy itself keeping track of the relationship between all of the different names.” —David Clark

Fixpoint is an OS that treats **Computations** as first-class objects



open()
read()
write()
socket()
close()
gettimeofday()

Status quo | Fix



Answers
Computational Truths

ISN'T THAT WHAT
COMPUTERS DO?



Fix: externalizing network I/O in serverless computing

Yuhan Deng
Stanford University

Akshay Srivatsan
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Francis Chua
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Yasmine Mitchell
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Stanford University

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Fix's different types allow programs to describe complex access patterns to FIXPOINT. A pseudocode version of the Fix API is shown in Table 1; the exact implementation varies depending on the implementation language.

Function	Description
T read_blob (BlobObject)	Read a Blob into a variable.
Value[] read_tree (TreeObject)	Read a Tree into an array.
BlobObject create_blob (T)	Create a Blob from a variable.
TreeObject create_tree (Value[])	Create a Tree from an array.
Thunk application (Tree)	Apply a function (lazily).
Thunk identification (Value)	Apply the identity function.
Thunk selection (Value, int)	Select a child element.
Encode strict (Thunk)	Strictly evaluate a Thunk.
Encode shallow (Thunk)	Shallowly evaluate a Thunk.

Table 1. Fix Pseudocode API

Fix's Thunks are lazy by default, which allows control flow to be expressed by user-provided programs. For example, Fig. 2 shows how a user-provided if procedure can lazily select one of two Thunks based on a predicate. The other

Algorithm 2 Fibonacci Procedure

```

[rlimit, fib, add, x] ← READ_TREE(input)
if x = 0 ∨ x = 1 then
    return CREATE_BLOB(x)
end if
x1 ← CREATE_BLOB(x - 1)
t1 ← CREATE_TREE([rlimit, fib, add, x1])
e1 ← STRICT(APPLICATION(t1))
x2 ← CREATE_BLOB(x - 2)
t2 ← CREATE_TREE([rlimit, fib, add, x2])
e2 ← STRICT(APPLICATION(t2))
tsum ← CREATE_TREE([rlimit, add, e1, e2])
return APPLICATION(tsum)
    
```

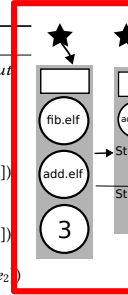


Figure 3. A Fix version of the Fibonacci algorithm that uses recursive Thunks and passes them to an addition procedure.

calls of itself, and a third Thunk adding the two results.

DM with Yuhan

[_] [X]

You 10:14 AM

How did you make that figure?

Yuhan 10:15 AM

All data and scripts are in the repo.

<https://github.com/fix-project/eurosyst26.git>

[+]

Type a message...

[Send]

File Browser	
<code>/home/user/eurosys26</code> Makefile eurosys26.pdf eurosys26.tex scripts/ - plot-add.py - plot-add-new.py graph/ - add.pdf	

File Browser	eurosys26.tex
/home/user/eurosys26 Makefile eurosys26.pdf eurosys26.tex scripts/ ├─ plot-add.py ├─ plot-add-new.py graph/ ├─ add.pdf	\includegraphics{graph/add.pdf}

File Browser	eurosys26.tex	
/home/user/eurosys26	<code>\includegraphics{graph/add.pdf}</code>	
Makefile	Makefile	DM with Yuhan [_] [X]
eurosys26.pdf		
eurosys26.tex	graph/add.pdf	You 10:17 AM
scripts/	python scri	So...plot-add.py?
└─ plot-add.py		
└─ plot-add-new.py	eurosys26.pdf	Yuhan 10:25 AM
graph/	pdflatex eu	Ummm actually I made some small tweaks and used plot-add-new.py at one point
└─ add.pdf	all: eurosys2	I'm not sure...
		🤔 😊 🌹 💀
		[+] Type a message... [Send]



SUM		fx	=A1 + B1
	A	B	
1	1	2	
2			= A1 + B1

make

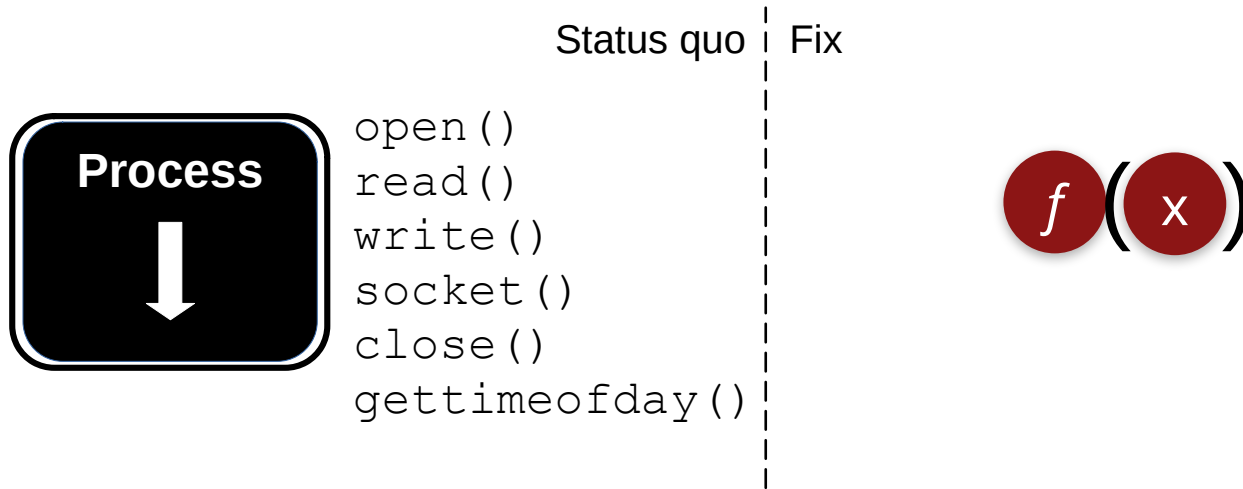
File Browser	Why
<pre>/home/user/eurosys26 Makefile eurosys26.pdf eurosys26.tex scripts/ ├─ plot-add.py └─ plot-add-new.py graph/ └─ add.pdf /usr/bin/ python pdflatex</pre>	
10 items	

File Browser	Why
<pre> /home/user/eurosys26 Makefile eurosys26.pdf eurosys26.tex scrip/ ├─ plot-add.py ├─ plot-add-new.py graph/ ├─ add.pdf /usr/bin python pdflatex </pre>	<pre> add.pdf ≡ /usr/bin/python plot-add-new.py </pre>
10 items • selected: add.pdf	

File Browser	Why
/home/user/eurosys26 Makefile eurosys26.pdf eurosys26.tex scripts/ └─ plot-add.py └─ plot-add-new.py graph/ └─ add.pdf /usr/bin python pdflatex	eurosys26.pdf ≡ /usr/bin/pdflatex "eurosys26.tex" /home/user/eurosys26
10 items • selected: eurosys26.pdf	

File Browser	Why
<pre> /home/user/eurosys26 Makefile eurosys26.pdf eurosys26.tex scripts/ ├─ plot-add.py ├─ plot-add-new.py graph/ ├─ add.pdf /usr/bin python pdflatex </pre>	<i>eurosys26.pdf</i> is equivalent to <i>pdflatex</i> (<i>python</i> (<i>plot-add-new.py</i>), <i>eurosys26.tex</i>) Computational Truth Reproducible by Anyone
10 items • selected: eurosys26.pdf	

Fixpoint is an OS that treats **Computations** as first-class objects



Serverless services

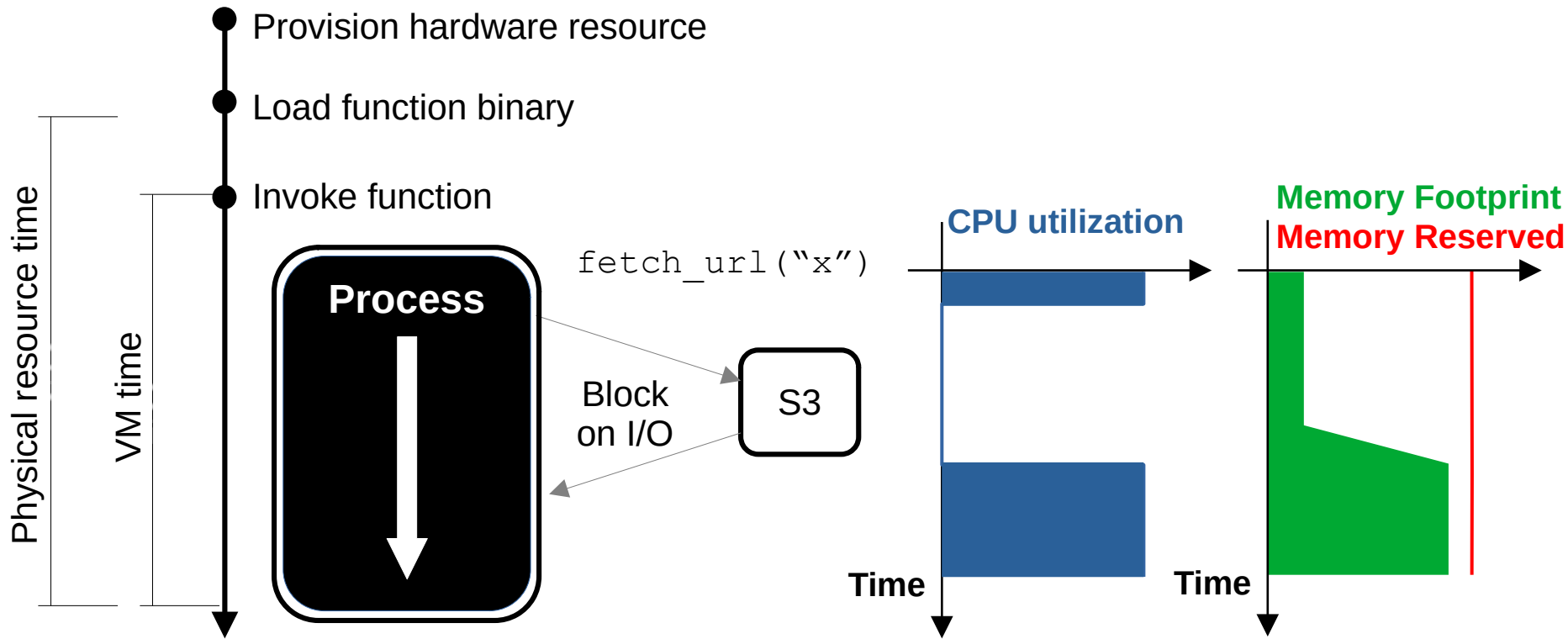


Thumbnailer.
zip

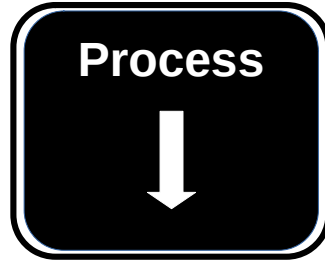
Function

`{"url": "x"}`

Input



User functions as processes provide no visibility



```
fetch_url("x")
```

1. Simultaneously transfer f and x

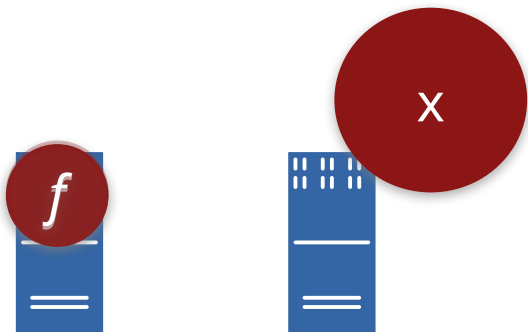


2. Delay allocating physical resources



Allocate vCPUs
and RAM

3. Choose a server close to the bigger dependency

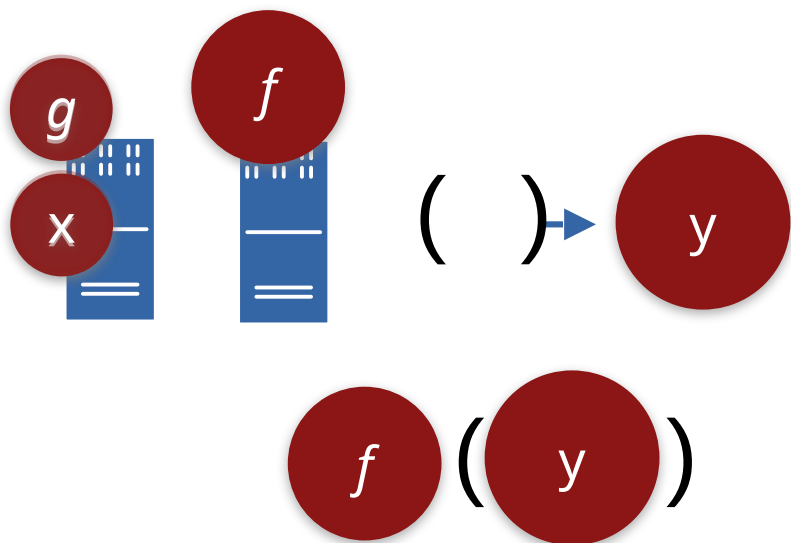


4. Delay invocation for batching opportunities

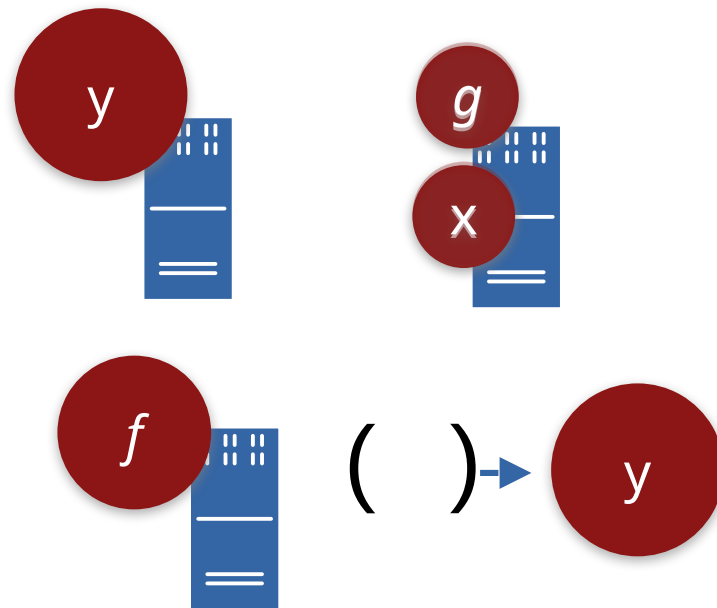


$$f(g(x))$$

5. Move to downstream dependency

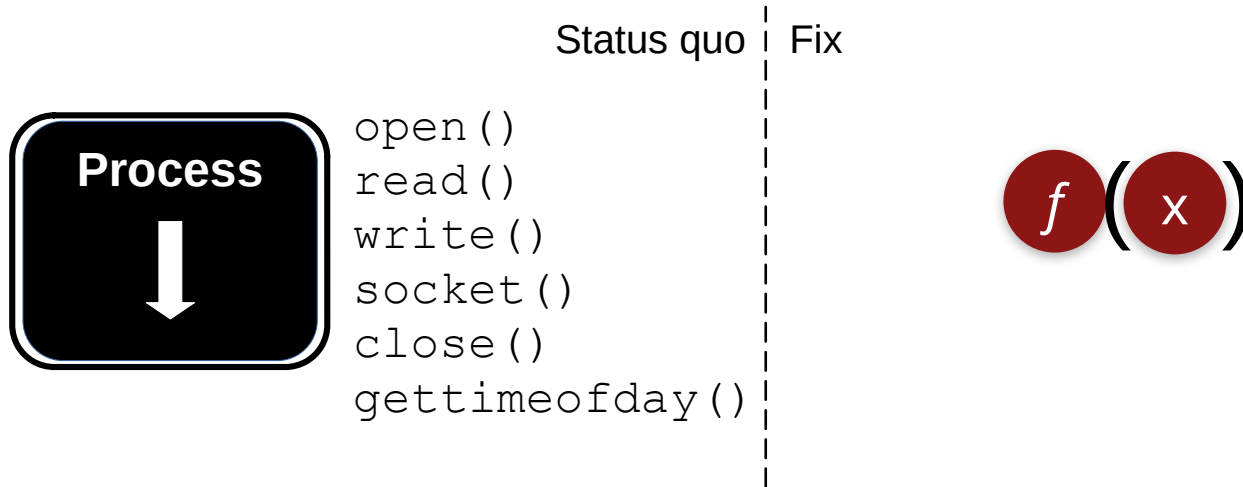


6. Recompute instead of moving large data



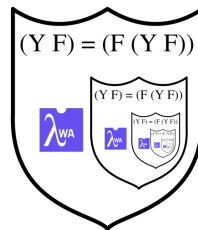
Fixpoint is an OS that treats **Computations** as first-class objects

With a common ABI Fix, shared by applications and infrastructure, to describe computation on data



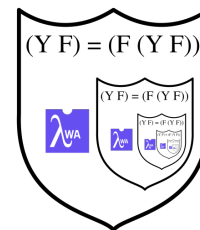
Outline

- Introduction
- Fix: a low-level ABI for computation
- Fixpoint: a Fix runtime
- Evaluation



Outline

- Introduction
- **Fix: a low-level ABI for computation**
- Fixpoint: a Fix runtime
- Evaluation



Fix ABI

Data

- **Blob:** an array of bytes
- **Tree:** a collection of Handles

Handles: identifications of named Data

Blob
Object



Tree
Object



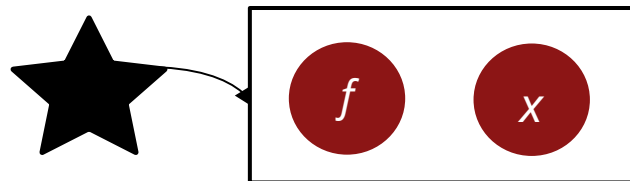
- **Thunk:** a computation

Application

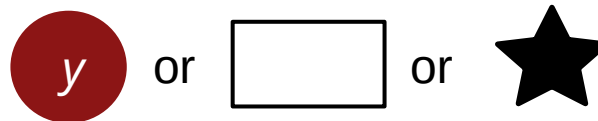


Each handle has a particular binary representation

User procedures are deterministic given specified dependency



computes to



Fix ABI

Data

- **Blob:** an array of bytes
- **Tree:** a collection of Handles

Handles: identifications of named Data

Blob
Object



Tree
Object

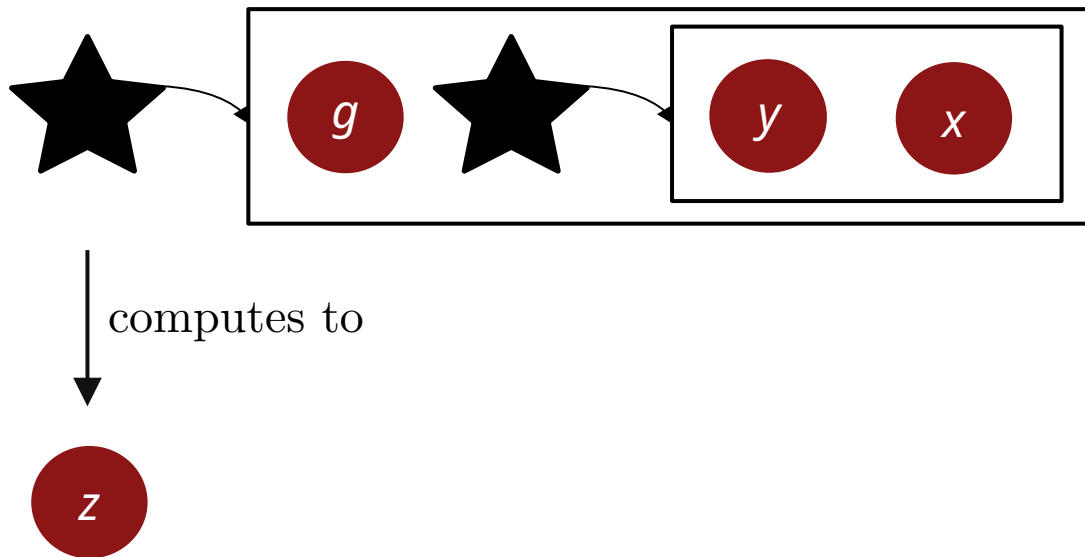


- **Thunk:** a computation
Application



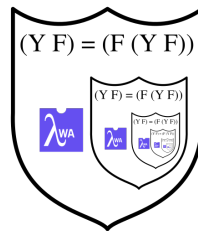
Each handle has a particular binary representation

$g(f(x))$



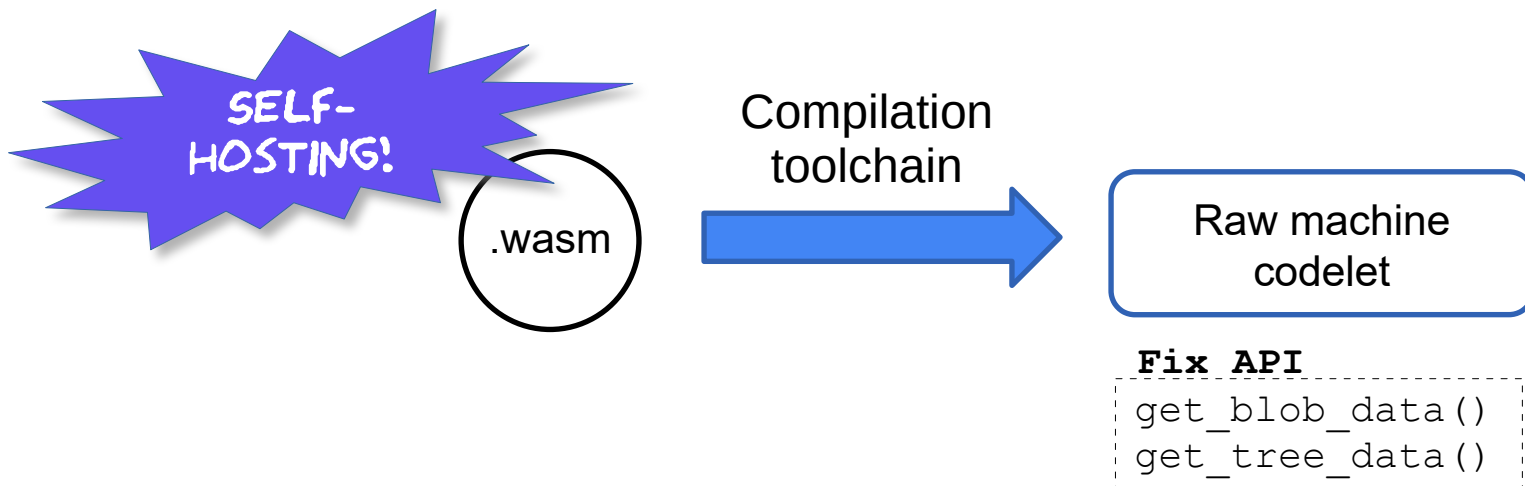
Outline

- Introduction
- Fix: a low-level ABI for externalizing I/O
- **Fixpoint: a Fix runtime**
- Evaluation



User procedures

User procedures are requested to be **memory-safe + deterministic**



add, Fibonacci, CPython, libclang, liblld, ...

 [mosh] yuhan@puffer5: ~/fixpoint

yuhan@puffer5:~/fixpoint\$./build/src/tester/fix eval application: tree:4 tree:1 uint64:1000000 compile: file:build/testing/wasm-examples/add-simple.wasm uint32:9 uint32:7 | xargs ./build/src/tester/fix decode

[mosh] yuhan@puffer5: ~/fixpoint

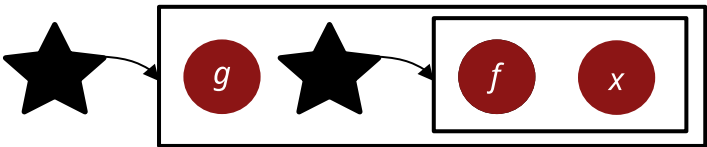
[mosh] yuhan@puffer5: ~

Fixpoint: a Fix runtime

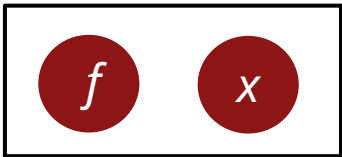
Fixpoint (Node 0)

Evaluator/Scheduler

- Locality-aware scheduling

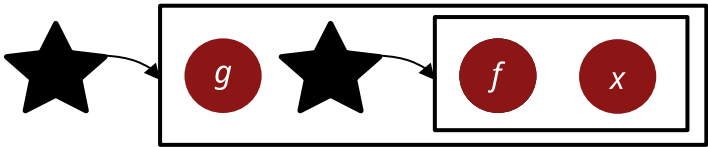


Lightweight function runtime (1.5μs)



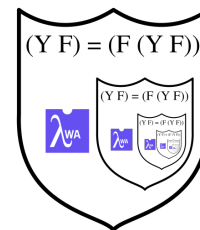
Raw bytes

Fixpoint (Node 1)



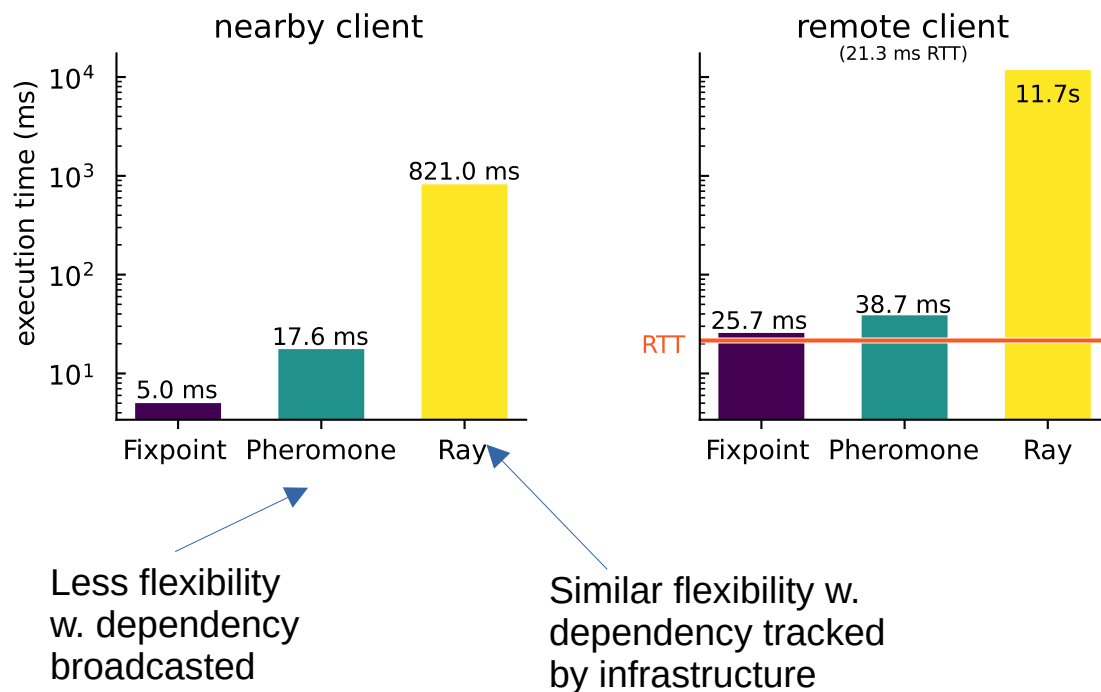
Outline

- Introduction
- Fix: a low-level ABI for computation
- Fixpoint: a Fix runtime
- **Evaluation**
- Coupon Collector



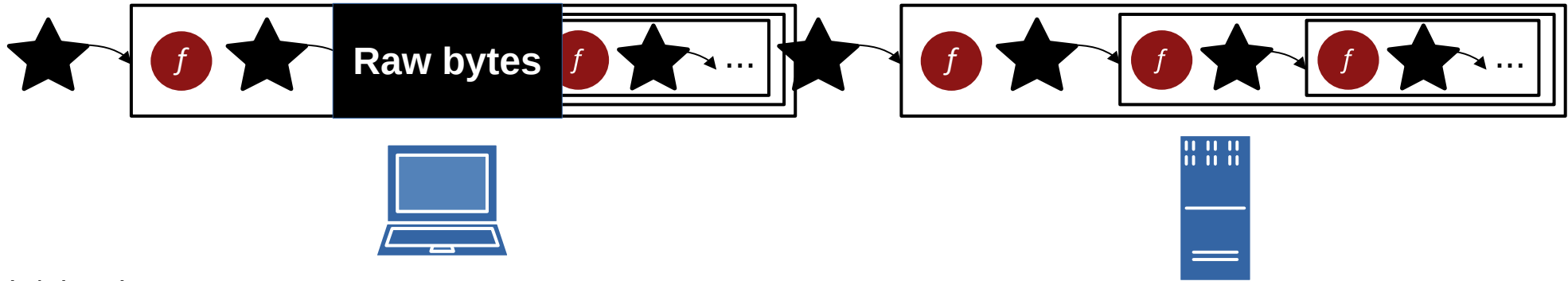
Invoking a chain of 500 function invocations

$$f(f(f(f(\dots))))$$



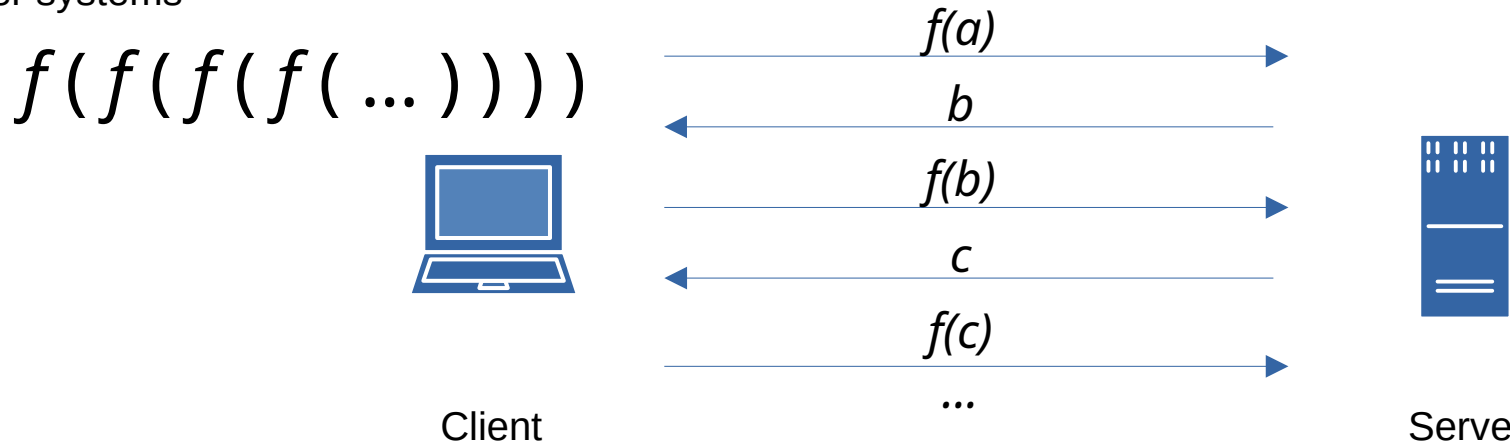
Invoking a chain of 500 function invocations

$$f(f(f(f(\dots))))$$



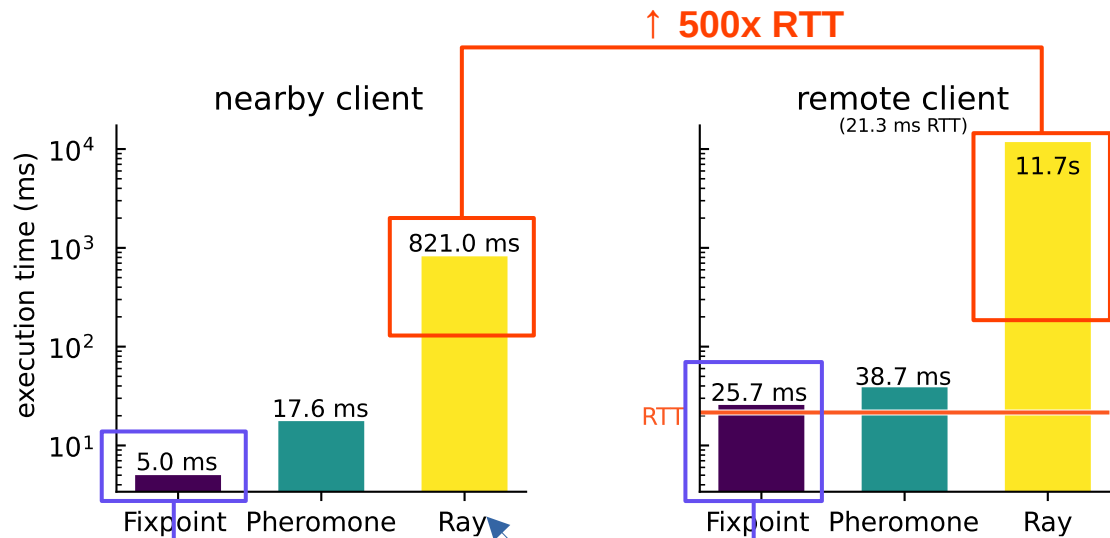
Fix/Fixpoint

Other systems



Invoking a chain of 500 function invocations

**Treating computations as first-class objects
avoids round-trips for resolving function dependency**

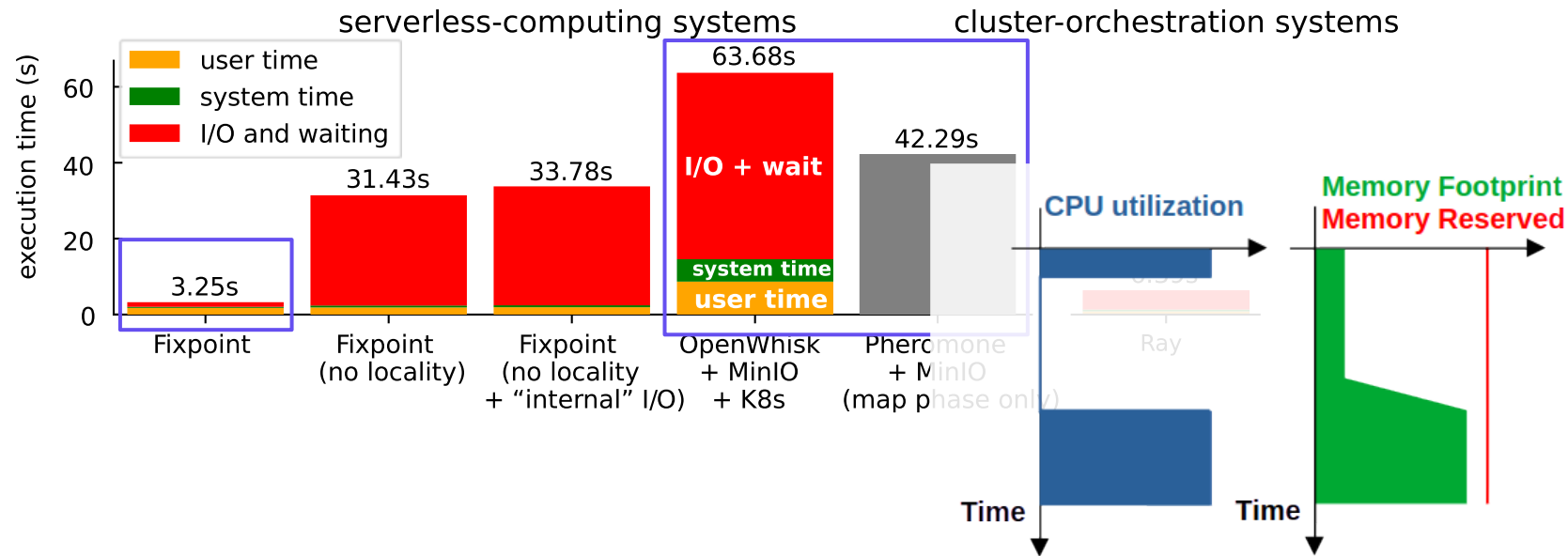


Less flexibility
w. dependency
broadcasted

Similar flexibility with
dependency tracked
by infrastructure

Counting occurrences of a string in Wikipedia dump

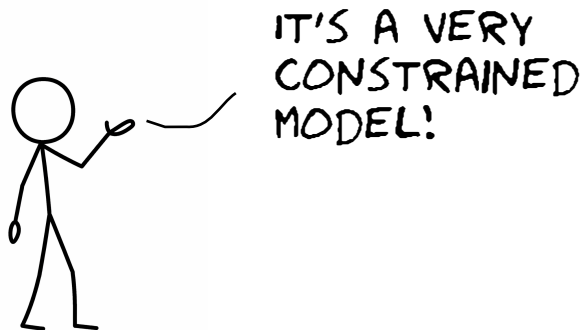
**Treating computations as first-class objects
allows locality-aware scheduling decisions**



Related Work

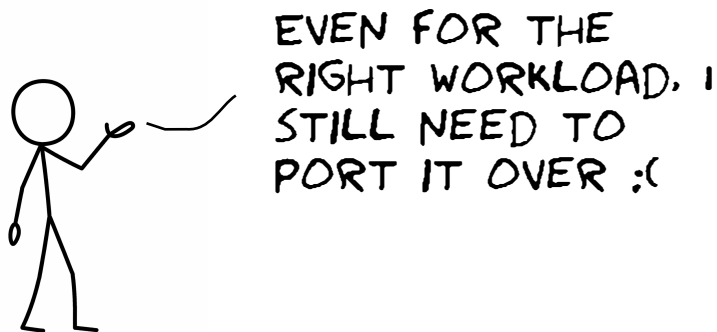
- ExCamera/gg: outsourcing computations to serverless nodes
- Docker/Nix: snapshotting file system dependencies
- Git: Blobs, Trees, and tracking the commit history of a repository
- CIEL/Ray: dependency model
- Dandelion: I/O-compute separation

Limitations



Functional workload only

- No shared mutable state
- No time measurement



Delineate nondeterminism

- get random seed

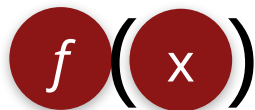
Recompilation

Make the Fix objects

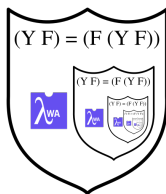
- e.g. to represent a file system

Fixpoint is an OS that treats **Computations** as first-class objects

With a common ABI Fix, shared by applications and infrastructure, to describe computation on data



Low overhead
Visibility for better scheduling
Reproducibility



github.com/fix-project

Contact: yuhan.deng@cs.stanford.edu